

COMP4450: Advanced Computing Research Proposal

Quantization-Induced SNR Degradation: Toward Signal-Preserving TinyML for Gravitational Wave Triggers

Razeen Wasif (u7283652)

Cluster

Intelligent Systems. This cluster is the ideal home, as my research focuses on optimizing machine learning algorithms for autonomous edge devices.

Keywords

TinyML Optimization, SNR Robustness, Network Quantization.

National Interest Test Statement

Australia is currently at the forefront of the global gravitational wave (GW) discovery effort through the ARC Centre of Excellence for Gravitational Wave Discovery (OzGrav) and the ANU Center for Gravitational Astrophysics. As the industry moves towards CubeSat-based detectors and autonomous terrestrial arrays [1], the volume of data generated is beginning to exceed available downlink capacity. The bottleneck for the next decade of multi-messenger astronomy is the “Data Deluge”—the inability to process and transmit massive raw data streams from space-based or remote terrestrial sensors.

This research addresses this challenge by developing autonomous edge intelligence: low-power “trigger” algorithms that permit sub-watt, local processing of astrophysical events on cheap, consumer-grade hardware. This benefits Australia by reducing reliance on international cloud infrastructure and expensive supercomputing for initial detection, lowering the operational costs of the Australian Space Agency’s upcoming missions, and positioning Australian computing research as a key enabler for the international LISA mission.

Introduction

As gravitational-wave (GW) observatories transition to space-based networks (e.g., LISA), processing continuous high-frequency data demands ultra-low-power edge computing. This shifts the computational burden from GPU clusters to embedded microcontrollers bounded by strict SRAM and sub-watt power constraints [4–6]. Situated at the intersection of Computer Systems, Machine Learning, and Computational Astrophysics, this research bridges embedded AI and GW physics. While deep learning models currently match classical matched-filtering pipelines in GW detection, their reliance on high-precision (Float32) numerical representations is incompatible with TinyML deployments.

Compressing neural networks to 8-bit integers (INT8) saves memory but risks introducing distortions that destroy the phase-coherence of GW “chirp” signals, fatally degrading the Signal-to-Noise Ratio (SNR). Aiming to establish a practical framework for deploying real-time GW triggers on ultra-low-power devices, this research asks: To what extent does non-linear bit-width (INT8) reduction interfere with the phase-coherence required for reliable GW signal triggering?

Research Problem

Traditional ML research in astronomy optimizes for *Global Accuracy*, which is a poor metric for GW detection where the *False Alarm Rate (FAR)* and *SNR recovery* are paramount. Most existing ML models for GW detection are “black boxes” optimized for desktop GPUs [2–5], ignoring the physical constraints of embedded systems—specifically the 256KB–512KB RAM limits of low-spec micro-controllers [6–8]. Current TinyML literature focuses on “human-centric” data (audio/images), where quantization errors are often masked by the data’s inherent redundancy. Astrophysical strain data, however, is non-redundant and stochastic.

The core research question is: **To what extent does non-linear bit-width (INT8) reduction interferes with the phase-coherence required for GW signal triggering?**

This work reframes quantization not merely as a compression technique, but as a physical perturbation to the signal representation that may alter the detectability of astrophysical events. I argue that current “off-the-shelf” quantization methods are largely agnostic to astrophysical signal fidelity, prioritizing computational precision without considering astrophysical signal fidelity or power constraints. [2–5, 7]. This work represents an advance by testing a “Signal-Preserving” optimization framework, filling the gap between systems engineering and theoretical physics, and providing an empirical benchmark of sensitivity-vs-memory to move GW detection from the super-computer to the “Edge.”

Research Design

The research employs a **Multi-Objective Experimental Design** following a quantitative methodology divided into three stages:

- 1. Data Synthesis & Acquisition:** Processing public GWOSC data, specifically using the PyCBC library to generate mock GW “chirp” signals injected into real LIGO O3 noise. This allows for controlled SNR levels to test the limits of INT8 math alongside environmental “glitches.”
- 2. Optimization Tiers:** Three models will be compared: (A) Baseline Float32 1D-CNN (Teacher), (B) Naive Post-Training Quantized (PTQ) INT8 model, and (C) Quantization-Aware Trained (QAT) “Student” model using Hardware-Aware Distillation.
- 3. Evaluation & Custom Kernel Development:** Standard TinyML frameworks (like TensorFlow Lite Micro) are optimized for streaming audio or visual data and incur significant overhead when processing high-sample-rate (4096 Hz) continuous whitened strain data. To address this, custom C++ inference kernels will be developed and optimized specifically for the ANU Rosella architecture (ARM Cortex-M). The models will then be evaluated to profile instruction-set efficiency, RAM usage, and inference latency, prioritizing astrophysical metrics (SNR recovery and FAR) over traditional classification accuracy.

If strict INT8 quantization irreparably degrades phase-coherence, the framework will pivot to Mixed-Precision Quantization (MPQ), retaining INT16 for highly sensitive initial feature extractors.

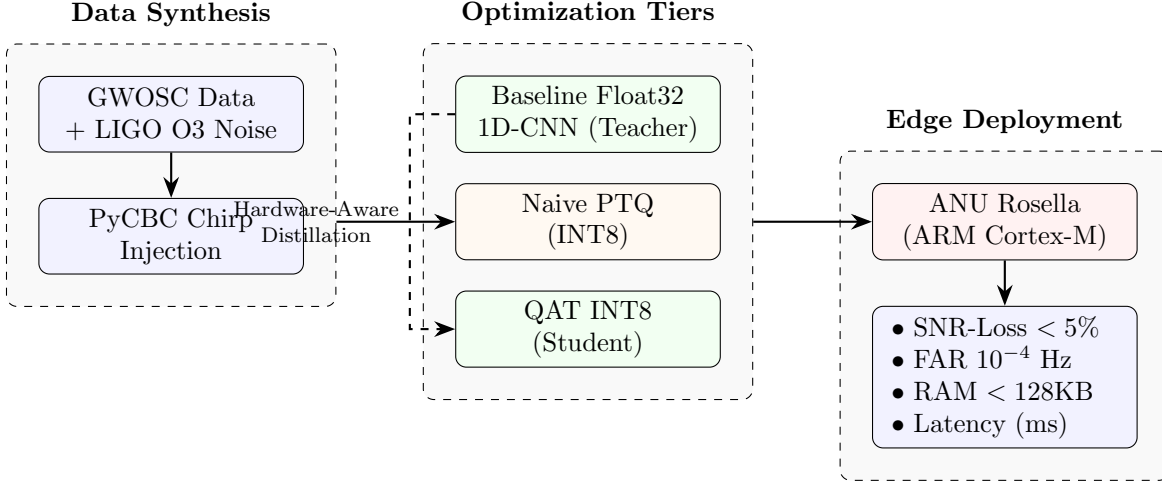


Figure 1: System architecture mapping the research design from mock chirp injection through the three-tiered quantization pipeline, culminating in hardware-in-the-loop evaluation on the ANU Rosella emulator against strict astrophysical (SNR, FAR) and computational (RAM, Latency) constraints.

Evaluation Criteria

The contribution will be validated through a comprehensive assessment of both astrophysical and computational metrics:

- **SNR-Loss Metric (Sensitivity):** The delta between the minimum SNR detectable by the baseline vs. the optimized model (targeting weak signals with $\text{SNR} < 8$) at a fixed False Alarm Rate of 10^{-4} Hz. Success is achieved if the QAT model maintains a sensitivity drop of $< 5\%$. This specific False Alarm Rate (10^{-4} Hz, or roughly one candidate trigger every 2.7 hours) is chosen as the optimal benchmark for an initial edge-level filter. While final astrophysical confirmation requires a much stricter FAR (e.g., $< 10^{-8}$ Hz), an edge-device operating at 10^{-4} Hz successfully compresses continuous data downlinks by over 99%, passing only highly probable candidate windows to the satellite bus or ground station for computationally heavy secondary confirmation.
- **Memory Efficiency:** Peak SRAM usage must achieve an $8\times$ reduction in memory footprint ($< 128\text{KB}$ SRAM).
- **Computational Latency:** Inference latency measured in milliseconds to ensure real-time viability.

Ethics

This study uses open-access, anonymized scientific data from the LIGO/Virgo collaboration. No human subjects or ethical conflicts are present, and ethical considerations are minimal.

Conclusion

This research will move gravitational-wave detection from post-hoc analysis to real-time edge detection. By demonstrating that GW triggers can function on low-power hardware, it provides a

blueprint for the next generation of autonomous astronomical sensors. It offers tangible benefits for researchers by enabling faster alerts and for satellite developers by optimizing limited on-board resources.

Reflection

The development of this proposal involved a critical reassessment of the capabilities of Generative AI (LLMs). While I used GenAI to assist in the LaTeX structural scaffolding and NIT plain-English synthesis, the core scientific “gap analysis”—specifically the tension between non-linear quantization and phase-coherence—required my independent synthesis of Physics and Computer Systems literature. I independently verified all domain-specific terminology (e.g., “chirp signal”) against the GWOSC documentation. This process enhanced my understanding of how academic integrity involves the critical verification of AI-generated abstractions against established physical constants.

References

- [1] Murphy et al., “A compact instrument for gamma-ray burst detection on a CubeSat (EIRSAT-1 GMOD),” *Journal of Astronomical Telescopes, Instruments, and Systems*, 2021.
- [2] George, D., & Huerta, E. A. (2018). Deep learning for real-time gravitational wave detection and parameter estimation with Advanced LIGO data. *Physics Letters B*, 778, 64–70.
- [3] Gabbard, H., Williams, M., Hayes, F., & Messenger, C. (2018). Matching matched filtering with deep networks for gravitational-wave astronomy. *Physical Review Letters*, 120(14), 141103.
- [4] Huerta, E. A., et al. (2022). Inference-optimized AI and high performance computing for gravitational wave detection at scale. *Nature Astronomy*, 6, 122–131.
- [5] Chua, A. J. K., et al. (2022). Topics at the interface of machine learning and high-performance computing for gravitational wave astrophysics. PhD thesis, University of Illinois at Urbana-Champaign.
- [6] Qi, H., Chen, H.-Y., Harms, J., Cusin, G., & Sesana, A. (2024). Improving early detection of gravitational waves from binary neutron stars using CNNs and FPGAs. *arXiv preprint arXiv:2409.05068*.
- [7] Gholami, A., Kim, S., Dong, Z., Yao, Z., Mahoney, M. W., & Keutzer, K. (2021). A survey of quantization methods for efficient neural network inference. *arXiv preprint arXiv:2103.13630*.
- [8] Banbury, C. R., et al. (2021). MLPerf Tiny benchmark: Measuring the performance of machine learning on microcontrollers. *Proceedings of Machine Learning and Systems (MLSys)*.
- [9] David, R., Duke, J., Jain, A., et al. (2021). TensorFlow Lite Micro: Embedded machine learning on TinyML systems. *Proceedings of Machine Learning and Systems (MLSys)*.
- [10] Cuoco, E., et al. (2024). Applications of machine learning in gravitational wave research with current interferometric detectors. *arXiv preprint arXiv:2412.19245*.
- [11] Gebhard, T. D., Kilbertus, N., Harry, I., & Schölkopf, B. (2019). Convolutional neural networks: A magic bullet for gravitational-wave detection? *Physical Review D*, 100(6), 063015.
- [12] Cuoco, E., Powell, J., Allweiss, S., et al. (2020). Enhancing gravitational-wave science with machine learning. *Machine Learning: Science and Technology*, 2(1), 011002.

Annotated Bibliography

George, D. and Huerta, E.A. (2018). Deep Learning for real-time gravitational wave detection and parameter estimation. *Physics Letters B*, 778, pp.64-70. <https://doi.org/10.1016/j.physletb.2017.12.053>

- Demonstrates that Convolutional Neural Networks can match classical matched-filtering sensitivity for binary black hole mergers. This is a seminal paper in GW machine learning, selected because its architecture provides the ideal Float32 ”Teacher” model baseline for my knowledge distillation methodology.
- While claiming ”real-time” processing, it relies on 250W desktop GPUs, ignoring the extreme sub-watt power constraints of embedded systems.

Abbott, B.P. et al. (2016). 'Characterization of transient noise in Advanced LIGO relevant to gravitational wave signal GW150914', *Classical and Quantum Gravity*, vol. 33, no. 13, p. 134001. <https://arxiv.org/abs/1602.03844>

- Details the exhaustive data quality vetoes used to rule out transient noise during the first GW detection, specifically "blip transients." I selected this foundational paper because it defines the exact morphological noise characteristics my edge-triggers must navigate to maintain strict False Alarm Rates.
- The paper relies on computationally heavy offline vetoes; it fails to address how real-time, ultra-low-power edge filters can achieve similar discrimination.

Gebhard, T.D. et al. (2019). Convolutional neural networks: a magic bullet for gravitational-wave detection? *Physical Review D*, 100(6), p.063015. <https://doi.org/10.1103/PhysRevD.100.063015>

- Critically examines CNN limitations in GW astronomy, proving they are highly effective as rapid "trigger generators" but vulnerable to adversarial noise. Selected because it validates my focus on edge-triggers and establishes the baseline fragility of neural networks facing non-standard noise.
- The study only evaluates high-precision models, failing to explore if INT8 mathematical truncation acts as a similar adversarial perturbation.

Gholami, A., et al. (2021). A Survey of Quantization Methods for Efficient Neural Network Inference, arXiv preprint, *arXiv:2103.13630*. <https://doi.org/10.48550/arXiv.2103.13630>

- Provides a comprehensive taxonomy of post-training quantization and quantization-aware training. This state-of-the-art survey was chosen because it outlines the exact mathematical framework—specifically Straight Through Estimators—required to mitigate gradient mismatch when compressing my models to INT8.
- It focuses entirely on human-centric data (images/text), failing to address quantization effects on high-frequency, non-redundant stochastic astrophysical strain data.

David, R., et al. (2021). TensorFlow Lite Micro: Embedded Machine Learning on TinyML Systems. *Proceedings of MLSys*. https://www.academia.edu/66158433/TensorFlow_Lite_Micro_Embedded_Machine_Learning_on_TinyML_Systems

- Details the strict runtime constraints and memory-arena planning required for bare-metal ARM Cortex-M microcontrollers. This is the industry-standard framework for TinyML and was selected because it defines the exact physical SRAM limitations my optimized model must adhere to.
- Optimized for streaming audio, it lacks support for high-sample-rate whitened strain data, necessitating the development of custom C++ inference kernels.